

user's head. The suits can be mobile or fixed/suspended according to requirements.

Solutions like Lokomat from Hocoma, a producer of robotic and sensor-based devices, give ergonomic support to the suffering people and allow the doctors to observe the patient's movements easily. Other companies making exoskeleton suits include Exo Bionics, ReWalk Robotics, Cyberdyne, Meditouch, and Bionik Laboratories.

Robotic exoskeleton from ReWalk has even received Food and Drug Administration (FDA) clearance for personal and rehabilitation use in the United States. It allows people with SCI to perform leg movements necessary for standing upright, walking, turning, and also using stairs.

Earlier, huge investment risks caused less manufacturers to tap this market. But these days, companies are offering fee-based, Robotics-as-a-Service pricing models for ensuring economic viability. Different types of exoskeletons are now available commercially. With a decrease in the cost of these wearables, hospitals can employ a wide variety of exosuits for meeting the needs of various patients.

With advancements in robotic technologies and growing demand from the healthcare sector for robotic rehabilitation, a number of companies are realising the benefits they can gain from the adoption of exoskeletons. At CES 2020 event in Las Vegas, Korean tech giant Samsung unveiled a fitness exoskeleton as well as a rolling robot that shows their interest in the healthcare field.

Leveraging the pros and dealing with the cons

There are numerous benefits of using exoskeletons in different fields. Several reports have findings that prove the safety of exoskeleton training, which indicates a likely positive growth of this field in future.

Exoskeletons make people less prone to fatigue resulting from per-

forming repetitive tasks for long durations, musculoskeletal disorders, or long-term injuries from accidents. Complex movements become easy with them because of jointed appendages.

These also prevent physical damage on the goods/materials side. Compared to previously existing options in the market, exoskeletons offer much more flexibility in work.

But at the same time, there are numerous problems too. Their structural design needs to accommodate several parameters and is difficult to make. Another problem is that their sometimes bulky and rigid structure can make them uncomfortable as a wearable. The manufacturers also need to fulfill stringent government regulations pertaining to medical applications.

In older designs, powering the exoskeletons required workers to be tied to a power source. But now, designs can be powered by small battery units that provide sufficient charging for long durations. Since the technology is still evolving with worldwide prototype tests going on, the use of next-gen exoskeletons in critical applications still has a long way to go.

Growth in India

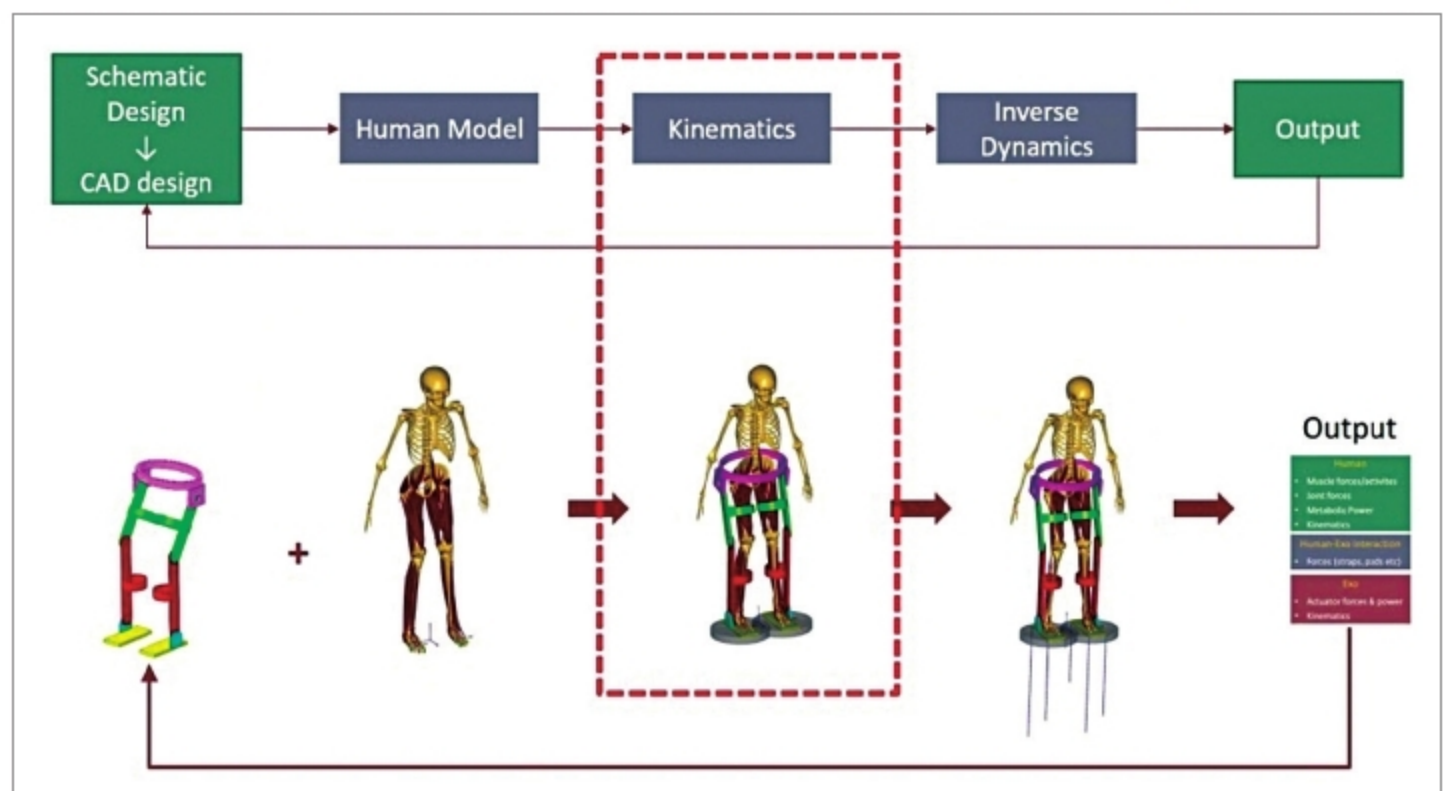
India has also seen an increasing number of companies and startups in this field, especially in the last five years. In 2015, ReWalk Robotics

announced a partnership with Saimed Innovations, a medical technologies provider, as part of an effort to distribute their exoskeleton systems in the country.

Being manufactured with a focus on Indian customers, the exoskeletons from Indian healthcare startup GenElek can do all the necessary work and are available at a low cost. Their technology works towards strengthening limb and hand movements for the rehabilitation of patients. It aids in reducing the recovery time too.

In February 2020, at Future Decoded tech summit, Microsoft showcased Bionic Yantra as one of the innovative startups on its cloud platform. The Indian startup developed an exoskeleton for people with spinal injuries to help them move their limbs in controlled conditions. With the assistance of the Internet of Things (IoT) and machine learning (ML), the data can be used for keeping a check on progress achieved and predicting patient recovery.

Research is ongoing to develop next-gen exosuits by both private and government sectors, with labs of institutions like Defence Research and Development Organisation (DRDO)'s Defence Bioengineering and Electromedical Laboratory (DEBEL) at the forefront. This is not just essential for adopting technology as a tool to address social needs, but also for boosting the Indian economy. **EFY**



Design process of human-centred exoskeleton (Credit: www.youtube.com)